

Deep Literature Review & Research Direction Task

Neurological Disorder Classification Using Signal Processing + Graph Neural Networks

We are developing a research project for **neurological disorder classification using biomedical/neurophysiological signals + signal processing + Graph Neural Networks (GNNs)**.

Our central research idea is:

Existing research has achieved neurological disease classification using traditional ML, deep learning, CNNs, RNNs/LSTMs, Transformers, handcrafted features, or signal-processing approaches, but many approaches do not model the relationships between signal-derived features, channels, brain regions, electrodes, frequency bands, or subjects as a graph.

We want to investigate whether a **Graph Neural Network can improve neurological disorder classification**, while also performing a rigorous comparison of **different signal preprocessing / feature-extraction pipelines on the exact same dataset and using the same downstream GNN architecture**.

The literature review should therefore identify the most suitable neurological disease, dataset(s), preprocessing pipelines, and research gap.

1. Primary Objective

Conduct a **deep literature review of at least 50 research papers** related to:

- neurological disorder classification
- EEG / electroencephalography
- EEG signal processing
- neurophysiological signal classification
- neurological disease detection
- Alzheimer's disease
- Parkinson's disease
- epilepsy
- dementia / MCI
- other relevant neurological disorders

- EEG-based disease diagnosis
- graph-based neurological analysis
- deep learning for neurological signal classification
- signal preprocessing for neurological disease classification

From these 50+ papers, identify and select the **30 most relevant papers** for our final literature review.

The final 30 should be highly relevant to our intended research direction rather than simply being papers containing the keywords "EEG" and "deep learning."

2. Most Important Selection Criterion: Existing GNN Research Gap

A major purpose of this literature review is to identify research where **GNNs/graph-based learning have not yet been used**, or where the authors themselves identify graph-based/GNN modeling as a future research direction.

Prioritize papers where the authors explicitly mention things such as:

- Graph Neural Networks
- GNN
- graph convolutional networks
- graph-based learning
- graph representation
- connectivity graphs
- brain graphs
- functional connectivity graphs
- electrode/channel relationship graphs
- network-based modeling
- spatial relationships between EEG channels
- graph convolution
- graph attention
- relational learning

as:

- future work
- a limitation
- a potential improvement
- an unexplored direction
- a missing methodological approach
- a recommendation for future research

VERY IMPORTANT

Do **not** classify a paper as having "GNN as future work" merely because you think a GNN could be applied to it.

For every selected paper, provide evidence showing why it qualifies.

Create a field:

"GNN Future-Work Evidence"

and provide:

1. Exact section of the paper where this appears.
2. Page number if available.
3. Short quotation or accurately paraphrased statement.
4. Explanation of how this supports our proposed research direction.

If the paper does not explicitly mention GNN/graph-based modeling but there is a strong methodological gap that clearly supports graph modeling, put it in a separate category:

"Potential GNN Opportunity — Not Explicitly Proposed by Authors"

Do NOT mix these two categories.

3. Read 50+, Select 30

First discover and examine **at least 50 potentially relevant papers**.

Then score and rank them.

Select the final **30 strongest papers**.

For every paper considered, record:

- Paper title
- Authors
- Year
- Journal/conference
- Publisher
- DOI
- DOI URL
- Dataset used
- Dataset size
- Disease studied
- Signal type

- Classification task
- Number of classes
- Preprocessing pipeline
- Feature extraction
- ML/DL model
- Evaluation metrics
- Best reported result
- Main limitation
- Future work
- Whether GNN/graph methods are mentioned
- GNN future-work evidence
- Citation count
- Citation source
- Relevance score

Also maintain a separate **"Papers Seen / Excluded"** table so that the literature-search process is reproducible.

4. Publication Quality Requirements

Prioritize papers from credible peer-reviewed sources such as:

- IEEE
- Springer
- Elsevier
- Nature / Scientific Reports
- Frontiers
- ACM
- Wiley
- MDPI only when scientifically strong and highly relevant
- major medical/neurology journals
- major ML/AI conferences
- established biomedical/neuroscience journals

Avoid:

- random blogs
- non-peer-reviewed articles
- low-quality journals
- predatory journals
- papers with unverifiable DOI
- papers where the dataset/results cannot be verified

Whenever possible, verify metadata using authoritative sources such as:

- DOI/Crossref
- PubMed
- IEEE Xplore
- Springer
- ScienceDirect
- Nature
- ACM Digital Library
- Google Scholar
- Semantic Scholar

Do not rely on a single website for citation counts.

5. Citation Count Requirement

We want papers with meaningful citation impact.

For each paper record:

- Current citation count
- Citation-count source
- Date citation count was checked

However, do NOT blindly rank a 2026 paper below a 2018 paper simply because the older paper has more citations.

Use two concepts:

A. Absolute citation impact

Raw citation count.

B. Citation velocity / age-normalized impact

Approximate citations per year since publication.

Use this especially when comparing:

- older highly cited papers
- recent 2025 papers
- recent 2026 papers

The final ranking should consider both **scientific relevance and citation impact**.

6. Mandatory 2025–2026 Requirement

At least:

15 of the final 30 papers must be published in 2025 or 2026.

The remaining 15 can be older foundational/high-impact papers.

Do NOT artificially include weak 2025/2026 papers merely to satisfy the quota.

If there are fewer than 15 genuinely relevant high-quality papers from 2025–2026, explicitly report this limitation instead of filling the list with irrelevant papers.

7. Dataset-Origin / Base Papers

For every dataset that becomes important to our proposed experiment, identify the:

original/base paper that introduced or established the dataset.

For example, if we ultimately choose a particular EEG dataset:

1. Find the original dataset publication.
2. Record its DOI.
3. Record dataset size.
4. Number of subjects.
5. Disease/control groups.
6. Recording protocol.
7. Sampling frequency.
8. Number of EEG channels.
9. Recording duration.
10. Available metadata.
11. Accessibility.
12. License/access requirements.
13. Number of subsequent publications using the dataset.

The dataset's original paper should be included in the literature database even if it is not one of the 30 disease-classification papers.

8. Disease Selection — VERY IMPORTANT

We have not finalized the neurological disease yet.

Analyze the literature and recommend the **best disease for our research project**.

At minimum investigate:

- Alzheimer's disease
- Mild Cognitive Impairment (MCI)
- Parkinson's disease
- Epilepsy
- Dementia
- other neurological disorders for which suitable signal datasets exist

Rank candidate diseases using:

Criterion	Weight
Public dataset availability	25%
Dataset size	15%
Difficulty / research challenge	15%
Existing SOTA performance	15%
Research gap / opportunity for GNN	15%
Clinical/scientific impact	10%
Reproducibility	5%

Give each disease a score from **1–10** for every criterion.

Then calculate an overall score.

Do not simply recommend Alzheimer's because it is popular.

We want the disease that gives us the strongest combination of:

dataset accessibility + sufficient data + meaningful classification problem + clear research gap + strong publication potential.

9. Dataset Ranking

For each candidate disease, identify the best publicly available datasets.

For each dataset report:

- Dataset name
- Official source
- DOI
- Dataset paper
- Number of subjects
- Healthy controls
- Disease subjects
- Disease subtypes
- EEG channels
- Sampling frequency
- Recording duration
- Signal format
- Data quality
- Missing data
- Artifact issues
- Accessibility
- License
- Whether preprocessing has already been standardized
- Number of papers using the dataset
- Whether train/test leakage is a known issue
- Whether subject-level splitting is possible

Rank datasets from best to worst for our project.

10. Signal → Numerical Representation / Preprocessing Literature

This is one of the MOST IMPORTANT parts of the research.

We want to compare different preprocessing pipelines while keeping:

- dataset
- train/test split
- labels
- downstream GNN
- evaluation metrics

as consistent as possible.

Find research papers using different ways of converting raw biomedical signals into numerical representations.

Investigate at least these categories:

Pipeline A — Raw/Minimal Processing

Raw EEG

- filtering
 - normalization
 - segmentation
 - numerical arrays
 - GNN
-

Pipeline B — Time-Domain Features

Raw EEG

- band-pass filtering
- artifact removal
- segmentation
- statistical/time-domain features
- feature vectors
- GNN

Potential features:

- mean
 - variance
 - RMS
 - standard deviation
 - skewness
 - kurtosis
 - zero-crossing rate
 - Hjorth parameters
 - entropy
-

Pipeline C — Frequency-Domain

Raw EEG

- filtering
- FFT / PSD
- frequency bands
- band-power features
- feature vectors
- GNN

Investigate:

- Delta
- Theta

- Alpha
 - Beta
 - Gamma
 - relative band power
 - spectral entropy
 - peak frequency
-

Pipeline D — Time-Frequency

Raw EEG

- preprocessing
- STFT / wavelet transform
- time-frequency representation
- numerical tensor
- GNN

Investigate:

- STFT
 - Continuous Wavelet Transform
 - Discrete Wavelet Transform
 - Wavelet Packet Transform
 - spectrograms
-

Pipeline E — Connectivity-Based

Raw EEG

- filtering
- source/channel signals
- functional connectivity
- connectivity matrix
- graph
- GNN

Investigate:

- correlation
- coherence
- phase-locking value
- phase lag index
- mutual information
- Granger causality
- functional connectivity

This pipeline is particularly important because the resulting connectivity matrix naturally maps to a graph.

Pipeline F — Time-Frequency + Connectivity

Raw EEG

- preprocessing
 - wavelet/STFT
 - frequency-band representation
 - connectivity calculation
 - graph construction
 - GNN
-

Pipeline G — Image-Like Representation

Raw EEG

- preprocessing
- spectrogram/scalogram
- 2D representation
- numerical tensor
- graph construction
- GNN

Investigate whether previous papers use:

- spectrograms
 - scalograms
 - topographic maps
 - recurrence plots
 - Gramian angular fields
 - channel-wise image representations
-

11. Build a Preprocessing Pipeline Comparison Matrix

For every relevant paper, identify its complete pipeline:

Raw signal → filtering → artifact removal → segmentation → transformation → feature extraction → normalization → final numerical representation → model

Create a comparison table.

Example:

Paper	Filtering	Artifact Removal	Windowing	Transform	Features	Representation	Model
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Then identify the **5–8 strongest preprocessing pipelines** that can realistically be implemented on the same dataset.

For each pipeline explain:

- Why it is scientifically valid
 - Computational cost
 - Information retained
 - Information discarded
 - Advantages
 - Disadvantages
 - Reproducibility
 - Compatibility with GNN
 - Risk of information leakage
 - Expected difficulty
-

12. Our Experimental Design

Based on the literature, propose a controlled experiment:

Independent Variable

Preprocessing / representation pipeline.

Controlled Variables

Keep constant:

- dataset
- subject split
- labels
- graph construction strategy where possible
- GNN architecture
- optimizer
- learning rate
- epochs
- evaluation metrics
- random seeds

Dependent Variables

Compare:

- Accuracy
- Precision
- Recall
- F1
- ROC-AUC
- PR-AUC
- Specificity
- Sensitivity
- Cohen's Kappa where appropriate
- confusion matrix

For imbalanced datasets, prioritize:

Macro-F1 + Balanced Accuracy + ROC-AUC + Sensitivity + Specificity

rather than accuracy alone.

13. GNN Architecture

We currently intend to use **one primary GNN architecture** so that preprocessing pipelines can be compared fairly.

Recommend the most appropriate architecture after examining the literature.

Evaluate candidates such as:

- GCN
- GAT
- GraphSAGE
- ChebNet
- GIN
- ST-GCN
- temporal GNNs if justified

Do NOT recommend an unnecessarily complicated architecture just because it gives higher SOTA numbers.

The primary experiment should favor:

simple + reproducible + explainable + appropriate to the graph structure.

If another architecture is substantially better, explain why.

14. Graph Construction

This is a critical research-design question.

Determine how the EEG should become a graph.

Investigate at least:

Option 1 — Electrode/Channel Graph

Nodes = EEG channels/electrodes

Edges = physical/spatial relationships between electrodes.

Option 2 — Functional Connectivity Graph

Nodes = EEG channels/regions

Edges = connectivity strength.

Option 3 — Frequency-Band Graph

Nodes = frequency bands/features

Edges = relationships between bands.

Option 4 — Feature Graph

Nodes = extracted signal features

Edges = statistical/semantic relationships.

Option 5 — Hybrid Graph

Nodes = channels/brain regions

Node features = extracted time/frequency/time-frequency features

Edges = functional connectivity.

Recommend the most scientifically defensible graph representation.

15. Binary vs Multiclass Classification

At the end of the literature review, give a strong recommendation:

Option A — Binary Classification

Disease vs Healthy Control

Example:

Alzheimer's vs Healthy

Option B — Multiclass Classification

Example:

Healthy
vs Alzheimer's
vs Parkinson's
vs another neurological disorder

Analyze:

- Dataset availability
- Biological validity
- Number of subjects
- Class balance
- Label reliability
- Clinical meaning
- Difficulty
- Existing SOTA
- Risk of confounding
- Publication potential
- Whether combining diseases from different datasets would create methodological problems

IMPORTANT

Do NOT recommend multiclass classification simply because it is harder.

Determine whether the classes are scientifically comparable and whether the same acquisition protocol/dataset supports the task.

If multiclass classification would require combining incompatible datasets, explicitly warn against it.

Give a final recommendation:

Binary / Multiclass / Hierarchical classification

and explain why.

16. Avoid Dataset Leakage

Pay particular attention to leakage.

Investigate whether papers incorrectly perform:

- window-level random splitting
- segment-level train/test splitting from the same patient
- preprocessing before splitting
- normalization using the entire dataset
- augmentation before splitting

Our preferred methodology should use:

subject-level train/validation/test splitting.

If possible, recommend:

- subject-independent evaluation
- stratified splitting
- cross-validation
- external validation

Explain why this matters for EEG.

17. Benchmarking Against Existing Research

For the selected disease/dataset, identify the strongest existing methods.

Create a SOTA table:

Paper	Year	Dataset	Task	Model	Preprocessin	Metric	Result
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Separate:

- Traditional ML
- CNN
- RNN/LSTM
- Transformer
- Hybrid models
- GNN/graph-based models, if any

Our proposed model should be compared fairly against the strongest relevant baselines.

18. Research Gap Statement

After reading the papers, formulate **3–5 concrete research gaps**.

Do not produce generic statements such as:

"More research is needed."

Instead identify precise gaps such as:

Existing approaches classify EEG-derived features independently or through convolutional architectures, while explicit relationships between EEG channels and functional connectivity are insufficiently modeled.

Or:

Existing studies use different preprocessing pipelines, making it difficult to determine whether performance improvements originate from the model architecture or from the signal representation.

Then determine which gap is:

Most novel + feasible + experimentally testable.

19. Proposed Contribution

Based on the literature, determine whether the following research contribution is defensible:

"A controlled comparison of multiple EEG signal preprocessing and representation pipelines using a common Graph Neural Network architecture for neurological disorder classification."

Then determine whether we can add:

"The study isolates the effect of signal representation from model architecture by holding the downstream GNN constant."

This distinction is important.

We do NOT want the paper to merely say:

"We used GNN for disease classification."

We want the paper to investigate:

How does the representation of neurological signals affect graph-based disease classification?

20. Required Final Output

Your final report must contain the following sections:

Section 1 — Executive Summary

Give us the answer in 1–2 pages:

- Best disease
 - Best dataset
 - Best classification task
 - Recommended preprocessing pipelines
 - Recommended GNN
 - Main research gap
 - Expected novelty
 - Main risks
-

Section 2 — Disease Ranking

Rank at least 5 candidate neurological diseases.

Include scoring and justification.

Section 3 — Dataset Ranking

Rank the best datasets for the selected disease.

Include dataset DOI and original dataset paper.

Section 4 — 50+ Paper Search

List all papers investigated.

Clearly mark:

- Selected
 - Rejected
 - Reason for rejection
-

Section 5 — Final 30 Papers

For every selected paper provide:

1. Citation
 2. Title
 3. Year
 4. Venue
 5. DOI
 6. Citation count
 7. Citation source
 8. Dataset
 9. Number of subjects
 10. Signal type
 11. Disease
 12. Classification task
 13. Preprocessing
 14. Features
 15. Model
 16. Results
 17. Limitations
 18. Future work
 19. GNN mentioned?
 20. GNN future-work evidence
 21. Relevance score
-

Section 6 — 2025–2026 Papers

Create a separate subsection containing at least 15 papers from:

2025 and 2026

Clearly mark them.

Section 7 — Dataset Papers

List the original papers introducing the datasets we may use.

Include DOI.

Section 8 — Preprocessing Pipeline Taxonomy

Create a complete taxonomy of:

Signal → Processing → Representation → Graph → GNN

Show all major approaches discovered in the literature.

Section 9 — Recommended Preprocessing Experiments

Recommend approximately **5–8 pipelines** that can be implemented on the same dataset.

Rank them.

Explain exactly how each converts:

raw signal → NumPy array/tensor → graph → GNN input

Include dimensions/shapes wherever possible.

For example:

Raw EEG

↓

Band-pass filter

↓

Epoching

↓

PSD

↓

Frequency-band features

↓
Node feature matrix X
↓
Adjacency matrix A
↓
GNN

Section 10 — GNN Recommendation

Recommend the primary GNN architecture and justify it scientifically.\

Section 11 — Graph Construction

Recommend:

- node definition
 - edge definition
 - node features
 - adjacency matrix
 - edge weights
 - graph normalization
-

Section 12 — Binary vs Multiclass

Give a final recommendation with evidence.

Section 13 — Experimental Protocol

Design the full experiment.

Include:

- preprocessing
- train/validation/test split
- subject-level splitting
- graph construction
- GNN
- hyperparameters

- evaluation
 - statistical testing
 - ablation studies
-

Section 14 — Proposed Research Questions

Generate 3–5 research questions.

Example:

RQ1: Which signal representation provides the strongest performance when evaluated using the same GNN?

RQ2: Does functional-connectivity-based graph construction outperform spatial electrode graphs?

RQ3: Does combining time-domain, frequency-domain and connectivity information improve neurological disorder classification?

RQ4: How sensitive is GNN performance to the preprocessing pipeline?

Improve these based on the literature.

Section 15 — Proposed Paper Contributions

Give 2–4 realistic contributions.

Separate:

Methodological contribution

Experimental contribution

Scientific contribution

Reproducibility contribution

Section 16 — Threats to Validity

Explicitly identify:

- dataset size

- class imbalance
- subject leakage
- preprocessing leakage
- overfitting
- site/device differences
- demographic confounding
- acquisition protocol differences
- hyperparameter sensitivity
- graph construction assumptions
- limited external validation

21. DOI Tracking — MANDATORY

Every research paper seen must have its DOI recorded whenever one exists.

Create a final machine-readable DOI list:

Paper 1 — DOI

Paper 2 — DOI

Paper 3 — DOI

...

Also provide a CSV-style table:

Paper	Year	DOI	DOI Verified	Selected
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Never invent a DOI.

If a paper genuinely has no DOI, explicitly write:

DOI: Not available

22. Evidence Standards

For every important claim, provide a source.

Especially verify:

- dataset size
- dataset availability

- reported accuracy
- preprocessing procedure
- model architecture
- limitations
- future work
- GNN mention
- DOI
- citation count

Do not infer experimental details when the paper does not provide them.

Use:

"Not reported"

instead of guessing.

23. Search Strategy

Use multiple search queries and databases.

Search combinations such as:

- EEG neurological disease classification
- EEG Alzheimer's classification
- EEG Parkinson's classification
- EEG epilepsy classification
- EEG preprocessing neurological disease
- EEG signal processing Alzheimer's
- EEG graph neural network future work
- EEG GNN limitations
- EEG graph convolution future work
- EEG functional connectivity disease classification
- EEG preprocessing comparison
- EEG representation learning neurological disorders
- neurological disorder classification deep learning limitations
- neurological disease EEG future work graph
- Alzheimer's EEG graph representation
- Parkinson EEG graph representation

Also search papers that do NOT use GNNs but have:

- channel relationships
- connectivity
- spatial information
- multivariate signals

- functional networks
 - limitations involving relationships between electrodes/brain regions
-

24. Important Distinction

Separate papers into these categories:

Category A — Explicit GNN Future Work

Authors explicitly mention GNN/graph-based learning as future work.

Category B — Graph-Based Future Work

Authors explicitly recommend graph/network/connectivity-based modeling but do not specifically say GNN.

Category C — Strong GNN Opportunity

Authors do not mention graphs/GNNs, but their limitations strongly motivate graph modeling.

Category D — Existing GNN

Papers that already use GNNs.

These are useful for understanding SOTA and graph construction, but they should NOT be counted toward the 30 papers required for the "GNN as future work" research-gap argument unless they independently satisfy the criterion.

25. Final Decision

At the very end, give us a concrete research recommendation in this format:

RECOMMENDED DISEASE:

...

RECOMMENDED DATASET:

...

RECOMMENDED TASK:

Binary / Multiclass / Hierarchical

RECOMMENDED GNN:

...

RECOMMENDED GRAPH:

...

RECOMMENDED PIPELINES:

- 1.
- 2.
- 3.
- 4.
- 5.

PRIMARY RESEARCH GAP:

...

PROPOSED NOVELTY:

...

EXPECTED BASELINES:

...

BIGGEST RISK:

...

GO / NO-GO:

...

Then answer this question directly:

If you were conducting this research yourself and wanted the highest probability of producing a scientifically credible and publishable paper, exactly what experiment would you run?

Give a concrete experimental design rather than a generic recommendation.

26. Critical Rule

Do not optimize this literature review for simply finding 30 papers.

Optimize it for finding a **defensible research gap**.

The final goal is to establish a chain of evidence:

Existing neurological classification research

↓

Different signal-processing/preprocessing approaches

↓

Limitations/inconsistencies in signal representation

↓

Insufficient modeling of relationships between channels/features/brain regions

↓

Opportunity for graph representation

↓

GNN as a unified downstream classifier

↓

Controlled preprocessing comparison

↓

Novel and reproducible experiment

The final report should make it possible for us to move directly from the literature review to:

**dataset selection → preprocessing implementation → graph construction → GNN
training → baseline comparison → ablation → statistical evaluation → research paper.**